

Shibam GHOSH

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I am a **Research Engineer** in the **COSMIQ** team at the French National Institute for Research in Digital Science and Technology (**INRIA, Paris**), working with **Léo Perrin**. I have completed my **PhD** thesis in the **Department of Computer Science** at the **University of Haifa**, under the supervision of **Prof. Orr Dunkelman**. I began my doctoral studies in October 2020. Before this, I completed my master's thesis as a research intern at **INRIA, Paris**, under the guidance of **Anne Canteaut** and **Léo Perrin**.

Research Interests

- Cryptanalysis of Symmetric-key Primitives
- Symmetric-key Primitive Design
- Lightweight Cryptography
- White-box Cryptography
- Provable Security

Education

Indian Statistical Institute

MASTER OF TECHNOLOGY IN CRYPTOLOGY AND SECURITY

Kolkata, India

2018 – 2020

Presidency University

MASTER OF SCIENCE IN MATHEMATICS

Kolkata, India

2015 – 2017

Krishnagar Government College

BACHELOR OF SCIENCE IN MATHEMATICS

Krishnagar, India

2012 – 2015

Research Experience

French National Institute for Research in Digital Science and Technology (INRIA)

RESEARCH ENGINEER

Paris, France

Jan 2025 – Currently

- Supervisor: **Léo Perrin**
- Focus Area: Design and Analysis of Symmetric-key Primitives for advanced cryptographic protocols.

Ethereum Foundation

RESEARCH INTERN

Remote

Oct 2024 – Dec 2024

- Supervisor: **Dmitry Khovratovich**
- Focus Area: Analysis of post-quantum digital signature schemes based of symmetric-key primitives

University of Haifa

PHD STUDENT

Haifa, Israel

Oct 2020 – Dec 2024

- Supervisor: **Prof. Orr Dunkelman**
- My PhD research has centered on cryptanalysis, focusing on the design and analysis of symmetric primitives.
- A significant portion of my work is dedicated to the algebraic cryptanalysis of these primitives. Some key contributions:
 - An enhanced Fast Fourier Transform-based key-recovery attack on 6-round **AES**.
 - Algebraic cryptanalysis of NIST LwC candidates **Ascon**, **KNOT**, and **TinyJAMBU**.
 - Distinguishers derived from the structural symmetry of **SHA3**, **Xoodyak**, and the Belarusian standard **bash**.
 - Division property-based fault attacks on the **GIFT** and **Present** ciphers.
- In addition to cryptanalysis, I have co-designed the tweakable block cipher **QARMAv2**, tailored for memory protection.
- My research also extends into the area of re-keying techniques, where I have analyzed the security of the IETF/ISO standard Advanced CryptoPro Key Meshing (**ACPKM**) internal rekeying technique, widely used in Russian variants of TLS and CMS.

- Supervisor: [Anne Canteaut](#), [Léo Perrin](#)
- Master's Dissertation: On the QIC of quadratic APN functions (available online [here](#))
- My master's research focused on cryptographic Boolean functions, with particular emphasis on the Big APN Problem.
- Our goal was to generate quadratic APN functions of size ≥ 8 .
- The search for APN permutations and their classification has been an open challenge for over 25 years.
- Our core approach involved representing a quadratic vectorial Boolean function using a cubic structure called a Quadratic Indicator Cube (QIC) and identifying the criteria associated with this cube that are necessary and sufficient for a function to be APN.

Publications

JOURNAL ARTICLES

1. Anup Kumar Kundu, Shibam Ghosh, Aikata Aikata, and Dhiman Saha. ToFA: Towards fault analysis of GIFT and GIFT-like ciphers leveraging truncated impossible differentials. *Cryptology ePrint Archive*, Paper 2024/1927, 2024 (To be appear in IACR CHES 2025)
2. Roberto Avanzi, Orr Dunkelman, and Shibam Ghosh. Differential cryptanalysis of the reduced pointer authentication code function used in arm's feat_pacqarma3 feature. *IACR Trans. Symmetric Cryptol.*, 2025(1):380–419, 2025 ([DOI](#))
3. Marcel Nageler, Shibam Ghosh, Marlene Jüttler, and Maria Eichlseder. Autodiver: Automatically verifying differential characteristics and learning key conditions. *IACR Trans. Symmetric Cryptol.*, 2025(1):471–514, 2025 ([DOI](#))
4. Sahiba Suryawanshi, Shibam Ghosh, Dhiman Saha, and Prathamesh Ram. Simple vs. vectorial: exploiting structural symmetry to beat the zerosum distinguisher. *Des. Codes Cryptogr.*, 93(1):133–174, 2025 ([DOI](#)).
5. Roberto Avanzi, Subhadeep Banik, Orr Dunkelman, Maria Eichlseder, Shibam Ghosh, Marcel Nageler, and Francesco Regazzoni. The QARMAv2 Family of Tweakable Block Ciphers. *IACR Transactions on Symmetric Cryptology*, 2023(3):25–73, 2023 ([DOI](#))
6. Orr Dunkelman, Shibam Ghosh, and Eran Lambooj. Practical Related-Key Forgery Attacks on Full-Round TinyJAMBU-192/256. *IACR Trans. Symmetric Cryptol.*, 2023(2):176–188, 2023 ([DOI](#))
7. Orr Dunkelman, Shibam Ghosh, and Eran Lambooj. Attacking the IETF/ISO Standard for Internal Re-keying CTR-ACPKM. *IACR Trans. Symmetric Cryptol.*, 2023(1):41–66, 2023 ([DOI](#))

CONFERENCE PAPERS

1. Yanis Belkheyar, Patrick Derbez, Shibam Ghosh, Gregor Leander, Silvia Mella, Léo Perrin, Shahram Rasoolzadeh, Lukas Stennes, Siwei Sun, Gilles Van Assche, and Damian Vizár. Chilow and chichi: New constructions for code encryption. In *EUROCRYPT*, volume 15601 of *Lecture Notes in Computer Science*, pages 212–243. Springer, 2025 ([DOI](#))
2. Orr Dunkelman, Shibam Ghosh, Nathan Keller, Gaëtan Leurent, Avichai Marmor, and Victor Mollimard. Partial Sums Meet FFT: Improved Attack on 6-Round AES. In *Advances in Cryptology – EUROCRYPT 2024*, pages 128–157. Springer, 2024 ([DOI](#))
3. Ravi Anand, Shibam Ghosh, Takanori Isobe, and Rentaro Shiba. Quantum Key Recovery Attacks on 4-Round Iterated Even-Mansour with Two Keys. In *Proceedings of Information Security Conference (ISC)*, volume 15257 of *Lecture Notes in Computer Science*, pages 87–103. Springer, 2024 ([DOI](#))
4. Shibam Ghosh, Anup Kumar Kundu, Mostafizar Rahman, and Dhiman Saha. Bizness: Bit invariant zero-sum property based on division trail. In *INDOCRYPT*, volume 15496 of *Lecture Notes in Computer Science*, pages 134–155. Springer, 2024 ([DOI](#))

5. Anup Kumar Kundu, Shibam Ghosh, Dhiman Saha, and Mostafizar Rahman. Divide and Rule: DiFA - Division Property Based Fault Attacks on PRESENT and GIFT. In *ACNS (1)*, volume 13905 of *Lecture Notes in Computer Science*, pages 89–116. Springer, 2023 ([DOI](#))
6. Orr Dunkelman, Shibam Ghosh, and Eran Lambooj. Full Round Zero-Sum Distinguishers on TinyJAMBU-128 and TinyJAMBU-192 Keyed-Permutation in the Known-Key Setting. In *INDOCRYPT*, volume 13774 of *Lecture Notes in Computer Science*, pages 349–372. Springer, 2022 ([DOI](#))
7. Nilanjan Datta, Avijit Dutta, and Shibam Ghosh. INT-RUP Security of SAEB and TinyJAMBU. In *INDOCRYPT*, volume 13774 of *Lecture Notes in Computer Science*, pages 146–170. Springer, 2022 ([DOI](#))
8. Shibam Ghosh and Orr Dunkelman. Automatic Search for Bit-Based Division Property. In *LATINCRYPT*, volume 12912 of *Lecture Notes in Computer Science*, pages 254–274. Springer, 2021 ([DOI](#))
9. Shibam Ghosh and Léo Perrin. Some Experimental Results on Quadratic APN Functions. In *Boolean Functions and their Applications (BFA) 2021*, 2021 (available online [here](#))

EPRINT ARCHIVE

1. Arghya Bhattacharjee, Ritam Bhaumik, Nilanjan Datta, Avijit Dutta, Shibam Ghosh, and Sougata Mandal. BBB secure arbitrary length tweak TBC from n-bit block ciphers. Cryptology ePrint Archive, Paper 2024/2049, 2024
2. Nilanjan Datta, Avijit Dutta, Shibam Ghosh, and Hrithik Nandi. Optimally secure TBC based accordion mode. Cryptology ePrint Archive, Paper 2024/2053, 2024

Research Talks Delivered

2025	Differential Cryptanalysis of the Arm's FEAT_PACQARMA3 Feature , FSE, 2025	Rome, Italy
2024	Partial Sums Meet FFT: Improved Attack on 6-Round AES , Eurocrypt, 2024	Zurich, Switzerland
2023	Attacking the IETF/ISO Standard for Internal Re-keying CTR-ACPKM , FSE, 2023	Kobe, Japan
2022	Full Round Zero-Sum Distinguishers on TinyJAMBU-128 and TinyJAMBU-192 , Indocrypt, 2022	Kolkata, India
2021	Automatic Search for Bit-based Division Property , Latincrypt, 2021	Virtual
2021	Some Experimental Results on Quadratic APN Functions , BFA, 2021	Virtual

Technical Projects

Improved Attack on 6-Round AES



PRACTICAL ATTACKS

- I have presented a Fast Fourier Transform-based key-recovery attack on 6-round **AES** at Eurocrypt 2024.
- The attack is fully implemented in C and verified on Amazon AWS servers.
- The source code is available in my [git](#) and published in IACR [artifact](#).

Forgery Attacks on NIST Lightweight Crypto Finalist TinyJAMBU



PRACTICAL ATTACKS

- I have implemented a practical related-key forgery attack on the **TinyJAMBU-v2** with 256/192-bit keys in C.
- Part of this work was published at FSE 2023 and the source code is available in my [git](#).

Automatic Tools For Algebraic Attacks



TOOLS IN CRYPTANALYSIS

- I have prepared Python-based automatic tools for cryptanalysis of NIST lightweight ciphers using MILP, SAT, and CP, available in my [git](#).

Automatic Tool for Verifying Differential Characteristics



TOOLS IN CRYPTANALYSIS

- This project involves developing a tool using publicly available #SAT solvers that thoroughly verifies differential characteristics.
- The tool calculates the expected probability of differential characteristics while considering the cipher's key schedule.
- This tool also estimates the size of the key-space that validates the characteristic and deduces conditions for these keys.
- The source code of the tool is available in [git](#).

Security Analysis of Arm's Pointer Authentication Code (PAC)



PRACTICAL ATTACKS

- This project focuses on the Qarma cipher, utilized in the Arm A-profile and M-profile architectures as a Pointer Authentication Code.
- Pointer Authentication Code (PAC) is important to mitigate Return-Oriented Programming (ROP) exploits.
- We have prepared automatic tools to search differential properties of Qarma and mounted practical attacks on various PAC versions.
- The source code of the project is available in my [git](#).

Ongoing Projects

White-Box Secure Cipher



PRIMITIVE DESIGN

- We aim to propose an Even-Mansour variant of white-box secure cipher family
- The main objective is to achieve space-hardness and longevity security in specific adversarial environments.

Teaching Experience

University of Haifa

Haifa, Israel

TEACHING ASSISTANT

2022 – 2024

- Introduction to cryptography, Spring Semester, 2022 (lecturer in charge: Prof. Orr Dunkelman)
- Introduction to cryptography, Spring Semester, 2023 (lecturer in charge: Prof. Orr Dunkelman)
- Introduction to cryptography, Spring Semester, 2024 (lecturer in charge: Prof. Orr Dunkelman)
- Israeli School on Biometrics, 2024 (lecturer in charge: Prof. Orr Dunkelman)

Other Professional Activities

REVIEWER

- Asiacrypt 2024
- Selected Areas in Cryptography (SAC) 2025, 2024
- Designs, Codes and Cryptography (DCC) 2025, 2024, 2023, 2022

Awards

2021	Israeli Science Foundation (ISF) Fellowship , Israeli Science Foundation (ISF)	Haifa, Israel
2020	Data Science Research Center (DSRC) Fellowship , University of Haifa	Haifa, Israel
2020	Awarded departmental-topper prize , Indian Statistical Institute	Kolkata, India
2019	Rank 88 in National Eligibility Test , Council of Scientific and Industrial Research (CSIR), India	Kolkata, India

Technical Skills



C, C++



python, SageMath



Rust



Git



Qiskit



Languages

Native Bengali

Fluent English, Hindi